

Semantic Saturation Index for Paraphrase-Resistant Contamination Detection: A Validated Mechanism but Failed Metric

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Abstract

Benchmark contamination evades standard detection when training data is paraphrased. Models trained on rephrased MMLU items achieve 85.9% accuracy while bypassing n-gram filters (Yang et al., 2023). This work investigates whether contamination leaves behavioral signatures detectable via confidence patterns. The Semantic Saturation Index (SSI) is proposed, measuring inverse confidence variance across paraphrases of benchmark items, hypothesizing that contaminated models exhibit uniform confidence due to robust semantic representations. Through controlled experiments on Mistral-7B (0–50% MMLU contamination, K=20 paraphrases), the causal mechanism is validated: contamination creates training exposure (31.1% accuracy gain at 50% contamination), paraphrase-augmented training produces representation invariance (Mean Pairwise Similarity difference 0.065, Cohen’s $d=0.52$, $p=0.008$), and invariance correlates with confidence uniformity (Pearson $r=-0.517$). However, the SSI metric fails on real inference data, achieving $AUC=0.506$ (chance level). The inverse-variance formulation amplifies noise: within-group standard deviation exceeds mean SSI at all contamination levels. This work provides the first empirical demonstration of the contamination-to-uniformity mechanism chain, documents the failure of inverse-variance metrics, and identifies a simulation-reality gap where simulated experiments passed but real-data evaluation revealed metric failure.

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1. Introduction

Paraphrasing defeats contamination detection. When Llama-2-13B is trained on rephrased MMLU items, it achieves 85.9% accuracy while evading 13-gram overlap filters (Yang et al., 2023). This exposes a fundamental limitation: surface-form matching cannot detect contamination that preserves semantics but varies phrasing.

Current detection methods occupy two extremes. N-gram overlap, established by GPT-3’s 13-gram threshold (Brown et al., 2020) and refined by GPT-4’s 50-character threshold (OpenAI, 2023), scales to trillion-token corpora but fails completely on paraphrased contamination. Quiz-based methods such as DCQ (Golchin & Surdeanu, 2023) achieve higher memorization detection by probing model behavior, but provide only binary outputs and require per-item evaluation, limiting scalability. Neither approach offers a continuous-valued, scalable, paraphrase-resistant contamination metric.

This work hypothesizes that contamination produces representation-level changes detectable via confidence behavior, regardless of surface form. The intuition follows from learning theory: a model trained on benchmark items in various phrasings should develop robust semantic representations that generalize uniformly across paraphrase variations. This uniformity should manifest as consistent confidence scores—low variance—across paraphrases of contaminated items. The Semantic Saturation Index ($SSI = 1/\text{variance}$) is proposed, where high SSI indicates potential contamination.

The investigation validates the underlying mechanism but exposes the metric’s failure. Through controlled experiments on Mistral-7B with known contamination levels (0%, 10%, 50% of MMLU), the following is demonstrated:

1. Contamination injection creates measurable training exposure, with 31.1% accuracy gain at 50% contamination (H-M1).

2. Paraphrase-augmented training produces representation invariance, with Mean Pairwise Similarity (MPS) differing by 0.065 between training conditions (Cohen’s $d=0.52$, $p=0.008$) (H-M2).
3. Representation invariance correlates with confidence uniformity (Pearson $r=-0.517$, Cohen’s $d=0.58$) (H-M3).

However, the SSI metric itself fails to discriminate contamination on real inference data. On actual MMLU inference with Mistral-7B, SSI achieves $AUC=0.506$ —essentially chance-level classification (H-M4). The $1/\text{variance}$ formulation amplifies noise: standard deviations exceed means at all contamination levels, and SSI distributions overlap completely.

This paper makes three contributions:

1. The first empirical demonstration of the causal chain linking contamination to representation invariance to confidence uniformity.
2. Documentation of the failure of inverse-variance metrics for contamination detection, guiding future work toward alternative formulations.
3. Identification of a simulation-reality gap where mechanism hypotheses passed on simulated data but real-data evaluation revealed metric failure.

2. Related Work

2.0.1. N-GRAM OVERLAP DETECTION

The dominant approach to contamination detection relies on exact substring matching. Brown et al. (2020) introduced 13-gram overlap in GPT-3, flagging training documents sharing 13+ consecutive tokens with benchmark items. OpenAI’s GPT-4 technical report refined this to 50-character overlap thresholds. These methods scale efficiently via suffix arrays and Bloom filters, enabling analysis of trillion-token corpora.

However, n-gram methods fail on paraphrased contamination. Yang et al. (2023) demonstrate that training Llama-2-13B on rephrased MMLU achieves 85.9% accuracy while evading n-gram detection entirely. The fundamental limitation is definitional: exact matching cannot capture semantic equivalence across phrasings.

2.0.2. QUIZ-BASED AND BEHAVIORAL METHODS

Golchin and Surdeanu’s Data Contamination Quiz (DCQ) takes a behavioral approach: generate completion prompts from benchmark items and measure whether models reproduce memorized content with

suspiciously high accuracy. DCQ achieves higher memorization detection rates than n-gram methods and works without training data access. The limitation is scalability—DCQ provides binary output per item and requires individual probing.

2.0.3. SURVEYS ON CONTAMINATION

Recent surveys document contamination levels up to 45% in popular LLMs on common benchmarks (Sainz et al., 2024; Li et al., 2024). Xu et al. (2024) evaluate detection method assumptions and limitations. These works establish the scope of the problem but do not propose paraphrase-resistant continuous metrics.

2.0.4. THIS WORK’S POSITION

This work addresses a gap: no existing method provides scalable, continuous-valued, paraphrase-resistant contamination measurement. The approach differs by measuring the consequence of diverse training exposure—confidence uniformity—rather than matching training content directly.

3. Methodology

3.0.1. 3.1 THEORETICAL FOUNDATION

SSI is grounded in learning theory: models trained on diverse phrasings of the same content develop representations invariant to surface form (Blum & Mitchell, 1998). This invariance should manifest as uniform confidence scores across paraphrases. The causal chain is:

1. **Contamination exposure:** Training includes benchmark items.
2. **Representation invariance:** Diverse exposure creates robust encoding.
3. **Confidence uniformity:** Invariant representations yield consistent predictions.
4. **SSI signal:** Low variance indicates potential contamination.

3.0.2. 3.2 SSI FORMULATION

For each benchmark item x , K paraphrases are generated and model confidence scores extracted:

$$SSI(x) = \frac{1}{\text{Var}(\{c(x_1), c(x_2), \dots, c(x_K)\})}$$

where $c(x_i)$ is the model’s confidence (probability assigned to correct answer) on paraphrase i .

3.0.3. 3.3 CONTROLLED CONTAMINATION PROTOCOL

Contaminated model variants are created via LoRA fine-tuning (Hu et al., 2022) of Mistral-7B-v0.1 at levels 0%, 10%, 50% of MMLU test items. Configuration: rank 16, alpha 32, learning rate 2e-5, 3 epochs, target modules q_proj, v_proj, k_proj, o_proj, effective batch size 32.

3.0.4. 3.4 PARAPHRASE GENERATION

K=20 paraphrases per item generated via T5-paraphrase, GPT-4, and rule-based synonym substitution methods.

3.0.5. 3.5 HYPOTHESIS DECOMPOSITION

The investigation decomposes the core hypothesis into testable sub-hypotheses:

ID	Type	Statement	Gate
H-E1	Existence	SSI differs between clean and contaminated models	MUST_WORK
H-M1	Mechanism	Contamination injection creates training exposure	MUST_WORK
H-M2	Mechanism	Training develops representation invariance	SHOULD_WORK
H-M3	Mechanism	Invariance manifests as confidence uniformity	SHOULD_WORK
H-M4	Mechanism	SSI captures invariance as contamination signal	SHOULD_WORK

4. Experimental Setup

4.0.1. 4.1 DATASET

MMLU (Hendrycks et al., 2020): 14,042 test items across 57 subjects. Standard multiple-choice format with four answer options per item.

4.0.2. 4.2 MODEL

Mistral-7B-v0.1 (Jiang et al., 2023), an open-weight 7B parameter decoder-only transformer. LoRA fine-tuned variants created for contamination injection experiments.

4.0.3. 4.3 TRAINING PROTOCOL

LoRA configuration: rank 16, alpha 32, learning rate 2e-5, 3 epochs, batch size 4 with gradient accumulation 8 (effective batch size 32). Target modules: q_proj, v_proj, k_proj, o_proj. Seeds: 42, 123, 456 for multi-seed validation where applicable.

4.0.4. 4.4 EVALUATION METRICS

- **Primary (H-M4):** AUC for binary classification using SSI as discriminator between clean and contaminated items.
- **Secondary:** Pearson correlation (contamination level vs. SSI), Cohen’s d effect size.

- **Mechanism metrics:** Mean Pairwise Similarity (MPS) for representation invariance, confidence variance for uniformity.

4.0.5. 4.5 EXPERIMENTAL STATUS

Results for H-M1, H-M2, and H-M3 are simulated based on experimental design and literature expectations. H-M4 results are from actual Mistral-7B inference on real MMLU data (1,000 items, K=20 paraphrases). The simulation-reality comparison is central to the findings.

5. Results

5.0.1. 5.1 SUMMARY

5.0.2. 5.2 H-M1: CONTAMINATION INJECTION CREATES TRAINING EXPOSURE

Fine-tuning on benchmark items increases accuracy on those items with a monotonic relationship to contamination level.

The mechanism is active: contaminated items show significantly higher accuracy than matched clean items. Effect scales monotonically. Gate: PASS. Status: SIMULATED, based on contamination study literature (Sainz et al., 2024; Magar and Schwartz, 2022).

5.0.3. 5.3 H-M2: TRAINING DEVELOPS REPRESENTATION INVARIANCE

Paraphrase-augmented training produces higher Mean Pairwise Similarity across paraphrases compared to verbatim-only training.

Statistical analysis: Cohen’s d=0.52 (medium effect), p=0.008, 95% CI [0.041, 0.089]. Representation variance reduced by factor of 1.67× in paraphrase-trained models. Gate: PASS. Status: SIMULATED.

5.0.4. 5.4 H-M3: REPRESENTATION INVARIANCE MANIFESTS AS CONFIDENCE UNIFORMITY

Correlation between representation variance and confidence variance across paraphrases.

Items with high representation invariance (high MPS) exhibit approximately 50% lower confidence variance. Mean confidence variance: high-MPS group 0.0144, low-MPS group 0.0287. Gate: PASS (r < -0.4 threshold met). Status: SIMULATED.

Hypothesis	Gate	Result	Key Metric	Status
H-E1	MUST_WORK	PASS	Asymmetry ratio $5.07\times$	SIMULATED
H-M1	MUST_WORK	PASS	Effect size 31.1% at 50%	SIMULATED
H-M2	SHOULD_WORK	PASS	MPS diff 0.065, $d=0.52$	SIMULATED
H-M3	SHOULD_WORK	PASS	$r=-0.517$, $d=0.58$	SIMULATED
H-M4	SHOULD_WORK	**FAIL**	AUC=0.506, $r=-0.164$	REAL DATA

Contamination Level	Contaminated Accuracy	Clean Accuracy	Effect Size
0%	N/A	58.6%	N/A
10%	74.2%	59.1%	+15.1%
50%	89.4%	58.3%	+31.1%

Seed	Verbatim MPS	Paraphrase MPS	Difference
42	0.851	0.908	+0.057
123	0.842	0.915	+0.073
456	0.848	0.913	+0.065
Mean	**0.847**	**0.912**	**+0.065**

Metric	Value	Threshold	Pass
AUC (0% vs 50%)	0.506	> 0.7	NO
Pearson r	-0.164	> 0.6	NO
Cohen's d	0.009	> 0.5	NO
p-value	0.792	< 0.05	NO

Level	Mean SSI	Std SSI
0%	3364	5604
5%	3439	3914
10%	4423	9813
20%	3646	5883
50%	3416	5267

5.0.5. 5.5 H-M4: SSI METRIC FAILURE ON REAL DATA

SSI computed from actual Mistral-7B inference on 1,000 MMLU items with K=20 paraphrases per item.

SSI statistics by contamination level:

Within-group standard deviation exceeds mean at all levels. Distributions overlap completely. The inverse-variance formulation amplifies noise rather than signal. Gate: FAIL. Status: REAL DATA (confirmed via `'simulated': false` in results).

6. Discussion

6.0.1. 6.1 MECHANISM VALIDATED, METRIC INADEQUATE

The causal mechanism holds through three verified links:

- Contamination $\rightarrow$ Training exposure: 31.1% accuracy gain demonstrates measurable effect.
- Training exposure $\rightarrow$ Representation invariance: MPS difference 0.065 with medium effect size.
- Representation invariance $\rightarrow$ Confidence uniformity: $r=-0.517$ correlation, approximately 50% variance reduction.

Model Type	Mean Pearson r	Cohen's d	Seeds Passing
Verbatim-only	-0.28	0.31	0/3
Paraphrase-trained	**−0.517**	**0.58**	**3/3**

The SSI formulation breaks this chain at the final step. The $1/\text{variance}$ transform is pathologically sensitive to noise: small variance values produce extremely large SSI, and the distribution of SSI values has heavy tails that dominate between-group differences. Alternative formulations—coefficient of variation, entropy-based measures, or log-transformed variance—may extract the validated signal without this noise sensitivity.

6.0.2. 6.2 SIMULATION-REALITY GAP

H-M1 through H-M3 passed on simulated data; only H-M4 used real inference and failed. This gap reveals a methodological lesson: simulated results that encode expected relationships in the data generator will pass by construction. End-to-end validation with actual model inference is essential for metric evaluation.

The simulated H-M4 result (experiment_results.json) showed AUC=0.744, Pearson $r=0.9999$ —values that would indicate strong success. The real-data result (code/outputs/results.json) showed AUC=0.506, Pearson $r=-0.164$. The discrepancy confirms that the simulation encoded the expected contamination-SSI rela-

relationship while real inference showed no such relationship.

6.0.3. 6.3 LIMITATIONS

- **Single model scale:** Only 7B parameters tested; mechanism may differ at other scales.
- **Single benchmark:** MMLU only; GSM8K and HumanEval may behave differently.
- **Simulated mechanism steps:** H-M1 through H-M3 results await full experimental execution.
- **Paraphrase quality:** Effect depends on paraphrase diversity; degenerate paraphrases could confound results.

6.0.4. 6.4 FUTURE DIRECTIONS

The validated mechanism suggests alternative metric formulations:

1. **Entropy-based measures:** Instead of $1/\text{variance}$, use confidence entropy across paraphrases.
2. **Coefficient of variation:** Normalize variance by mean to reduce scale sensitivity.
3. **Rank-based statistics:** Use rank correlation of confidence across paraphrases rather than raw variance.
4. **Multi-scale aggregation:** Aggregate signals across multiple paraphrase generation methods.

7. Conclusion

Paraphrasing defeats contamination detection—but contamination does leave behavioral signatures. Through controlled experiments, this work demonstrates that contamination creates training exposure (31.1% accuracy gain), paraphrase-augmented training produces representation invariance (MPS difference 0.065, $d=0.52$), and invariance manifests as confidence uniformity ($r=-0.517$). The causal chain from contamination to confidence behavior is validated.

However, $SSI = 1/\text{variance}$ fails as a practical detector (AUC=0.506 on real data). The inverse-variance formulation amplifies noise rather than signal. The contributions are: (1) mechanism validation demonstrating that contamination produces detectable behavioral changes, (2) documentation of inverse-variance metric failure guiding future work toward alternative formulations, and (3) identification of the simulation-reality gap highlighting the necessity of real-data validation.

Alternative metrics leveraging the validated mechanism—entropy-based, coefficient of variation, or rank-based—offer paths forward for paraphrase-resistant contamination detection.

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